

TABLE 1: IEEE 802.11 PHY STANDARDS

Release date	Standard	Band (GHz)	Bandwidth (MHz)	Modulation	Advanced antenna technologies	Maximum data rate
1997	802.11	2.4	20	DSSS, FHSS	N/A	2 Mbits/s
1999	802.11b	2.4	20	DSSS	N/A	11 Mbits/s
1999	802.11a	5	20	OFDM	N/A	54 Mbits/s
2003	802.11g	2.4	20	DSSS, OFDM	N/A	54 Mbits/s
2009	802.11n	2.4, 5	20, 40	OFDM	MIMO, up to four spatial streams	600 Mbits/s
2012 (expected)	802.11ad	60	2160	SC, OFDM	Beamforming	6.76 Gbits/s
2013 (expected)	802.11ac	5	40, 80, 160	OFDM	MIMO, MU-MIMO, up to eight spatial streams	6.93 Gbits/s

not that popular so offers consistent speeds. But it does not penetrate walls and other obstacles the same way the 2.4 GHz band does and thus offers a significant disadvantage as well. Typically, you should have good speeds till 30 meters, greater than that you should still be able to access but with a lot of signal drops it would hardly be worth the effort. Hybrid devices exist that use two bands at the same time to double up the transmission speeds.

WiFi routers these days adhere to a multitude of 802.11 protocol revisions, i.e., the same router supports 802.11b / 802.11g / 802.11n / 802.11ac. The primary difference is the bandwidth that they offer. The following table lists the same for each revision.

Now you need to understand that there are data streams allocated to each band. The typical annotation will be 1x1 or 2x2 or 3x3. This means that there are 'upload streams' x 'download streams'. Each device that connects to the device gets allocated a portion of the net bandwidth that the router can support. So when a router boasts that it can handle 900 Mbit/s it means that overall it can output that over your LAN but individually you might get somewhere around 130-150 Mbit/s.

WiMAX

A relatively new entrant, WiMAX or IEEE 802.16 is a technology that uses huge transmission antennae to provide internet access over a huge area, typically around 40-50 kilometers. It offers 1 Gigabit/s transmission rates for fixed stations. By fixed station, we mean a stationary receiver. Mobile users will suffer a significant reduction in transmission speed. 1 Gigabit/s